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|---------------------------------|------------------------------------------------------|
| Course- B.Tech.                 | Type- Core                                           |
| Course Code- <b>2025CSET106</b> | Course Name- <b>Discrete Mathematical Structures</b> |
| Year- I                         | Semester-II                                          |

## Tutorial-1

| Tutorial No. | Name                                     | CO1 | CO2 | CO3 |
|--------------|------------------------------------------|-----|-----|-----|
| <b>1</b>     | <b>Propositions, Logical Expressions</b> | ✓   | --  | --  |

**Objective:** The main objective of this tutorial is to learn about propositions, basic operations in propositions, translate English statements into logical expressions.

Q1. Consider the following six statements:

- Ice floats in water.
- $7 > 26$ .
- How far is Delhi from Greater Noida?
- $2 + 2 = 5$
- Oh no!
- It is false that  $2 + 2 = 5$ .
- $x$  is a prime number
- Tomorrow will be a great day.

Recognize the propositional statements. Also, state their truth values.

Q2. Suppose that during the last year, the average daily passenger counts and on-time performance of three metro lines were as follows:

- Red Line carried 1.8 million passengers per day and had 92% on-time performance.
- Blue Line carried 1.2 million passengers per day and had 88% on-time performance.
- Green Line carried 1.5 million passengers per day and had 95% on-time performance.

Determine the truth value of each of the following propositions.

- The Red Line carried the highest number of passengers, and the Green Line had the best on-time performance.
- The Blue Line carried the fewest passengers and either the Red Line or the Green Line had on-time performance above 90%.
- The Green Line did not carry the highest number of passengers, and neither the Red Line nor the Blue Line had the worst on-time performance.

Q3. For each of these sentences, determine whether an inclusive or, or an exclusive or, is intended. Explain your answer.

- Experience with C++ or Java is required.
- Dinner includes rice or bread.
- Flights are cancelled if visibility is poor or if wind speed exceeds 80 km/h..
- When you enroll in the new internet plan, you receive a free router or a 3-month subscription discount.

Q3. Determine whether the given compound propositions are tautology, contradiction or contingent

- a.  $(p \vee q) \wedge \neg (p \vee q)$
- b.  $(p \wedge q) \vee \neg r$
- c.  $(p \oplus q) \vee \neg (p \oplus q)$

Q4. Let p and q be the propositions

p: The library is open.

q: The librarian is available.

Write the following propositions using p, q, and logical connectives (including negations).

- a. The library is open but the librarian is not available.
- b. The library is not open, and the librarian is not available.
- c. Either the library is open or the librarian is available (or both).

Write the following propositions using p, q, and logical connectives (including negations).

Q5. Let p, q, and r be the propositions:

- p : You pass the fitness test.
- q : You attend all training sessions.
- r : You are selected for the team.

Write each statement using p, q, r, and basic logical connectives.

- a. You are selected for the team, but you do not attend all training sessions.
- b. You pass the fitness test, attend all training sessions, and are selected for the team.
- c. You pass the fitness test, but you do not attend all training sessions; nevertheless, you are selected for the team.

Q6. Let p, q, and r be the propositions

p : You have the flu.

q : You miss the final examination.

r : You pass the course.

Express each of these propositions as an English sentence.

- a.  $p \vee q \vee r$
- b.  $\neg p \wedge q$
- c.  $(p \wedge q) \vee (\neg q \wedge r)$